THIS PAGE CAN BE DELETED FROM YOUR FINAL REPORT

(ALL text in blue can be deleted from your final submission, Text in red should be edited by you)

The following guidelines **must** be adhered to:

Writing Style

Your report should usually be written from a third person, observational perspective. Do not use the following terms:

Do use or model from the following examples which are written from an observational third person perspective:

"The researcher" or "the observer"	"The procedure was executed as follows..."
"It was observed that..."	"The measurements were taken..."
"The data suggest..."	"The subjects/participants behaved in a certain way..."
"The experiment revealed..."	"The test conditions were controlled..."
"The results show..."	"The trends in the data"
"One can infer that..."	"The observations were made by the team."
"The study indicates..."	"The process was documented as..."
"The findings imply..."	"The phenomenon was recorded."
"The evidence supports..."	"A statistical analysis was conducted."
"An analysis of the data"	"The data were collected and analysed."

Report Layout and Style

The following guidelines **must** be adhered to:

- All **text will be black** in the report unless in very exceptional circumstance.
- Main body text must be at least **11pt font** using either **Arial or Calibri** font.
- Main body text will have **1.5 line spacing**.
- **Margins** will be a minimum of **2 cm on each side**.

- All **pages** will be **numbered consecutively**.
- Start **each section on a new page**.
- **Figures** must have **captions** and be **numbered** (e.g., Figure 1).
- **Tables** must have **captions** and be **numbered** (e.g., Table 1).
- **Figures** may be **black and white**, or **colour**.

Word Count

- There is a **8,000-word limit** for the project. This is a maximum, not a target. Shorter is often better.
- All words in the main body, excluding words in figures and tables, will be included in the word count.
- Consider what can be removed from the main body and placed in an appendix to reduce the word count as not every detail may be necessary in the main document.

Copyright and Intellectual Property Rights (IPR)

Your report should be written considering that it will be within the public domain. Normally, you retain copyright over your written work and Intellectual Property Rights (IPR) over any technical work. There are rare situations where this might not be as simple, for example when working with a company, external client, or on a larger university project. There are strategies you can utilise:

- You can inform your supervisory team that the project cannot be made available to other students undertaking an MSc Project.
- You can provide a shortened report for sharing that does not contain the protected information.
- You can assign IPR to the external collaborator, although you should take great care when doing so. It is best to speak to your supervisor.
- Non-disclosure agreements can be made between the external collaborator and the supervision team.

VOLog

<Subtitle if required>

By

Hala Bakhtiar

Submitted to

The University of Roehampton

In partial fulfilment of the requirements

for the degree of

BACHELOR OF SCIENCE/ENGINEERING IN SOFTWARE ENGINEERING

Declaration

I hereby certify that this report constitutes my own work, that where the language of others is used, quotation marks so indicate, and that appropriate credit is given where I have used the language, ideas, expressions, or writings of others.

I declare that this report describes the original work that has not been previously presented for the award of any other degree of any other institution.

Enter your name here

Date: Enter the date here

Signed (apply signature below)

Acknowledgements

Here it is customary to thank the people who have supported this work and your studies in general.

It is up to you who you thank!

Abstract

GUIDANCE: Up to 300 words

A short summary of your project to include the problem, the main literature reviewed, your implementation and your findings.

Write this after you have finished the entire report!

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1. Introduction

An initial paragraph introducing the project motivation, context and idea

Slightly generalise language and remove sensitive specifics.

Large outdoor gatherings spanning over 200 acres, with multiple access gates and partial fencing, pose significant challenges for monitoring entry and exit. While thousands of attendees occupy the main event space, certain vehicles, such as maintenance staff or authorised personnel, must access the inner fenced areas. Current systems relying on manual logs of vehicle registration numbers, arrival times, and passenger counts are fallible and time-consuming, especially when coordination across gates relies on limited radios or phone signals. These challenges amplify the risk of data inconsistencies, unauthorised entries, and delays in decision-making.

This project aims to develop an automated vision-based entry security system that detects the headcount in vehicles, records licence plates, and logs entry times digitally. By integrating computer vision with federated learning, the system can train models locally at each gate. The solution will provide real-time, centralised data, helping them manage multiple gates, maintain data privacy and improve accuracy.

Problem Statement and/or Research question

Your research question or problem statement.

This should include the following considerations:

- What is the problem you are addressing?
- Who is affected by the problem?
- Where and/or when does the problem occur?
- Why is it important to solve the problem?

Rewrite in third person, don't change conceptually.

Managing entry and exit across multiple gates in an outdoor field during a large gathering has proven difficult due to manual record-keeping and limited communication tools, such as radios. Currently, the logging of vehicle registration numbers, arrival times, and the number of passengers is written on paper. Coordination between these gates relies on radios, which are not consistently accessible to all staff members, making it difficult to maintain accurate records and real-time oversight.

This manual process leads to gaps in security, such as allowing unauthorised vehicles to enter the site, inconsistent counting of external workers who have entered and left the area, and handwriting errors. A vision-related computer system automatically detects the number of occupants, captures license plate details, and records the time of entry, which can significantly improve accuracy, efficiency, and coordination across all entry points.

Aims

Aims are broad statements of desired outcomes and intentions. We expect about 3.

Lecturers said don't use bullet points.

1. To design an automated system that records vehicle registration, time, and passenger count using camera-based detection.
2. To replace manual processes with a real-time, centralised digital system.
3. To improve security, team coordination, and monitoring efficiency at multi-gate outdoor field events.

Objectives

Objectives are the tangible steps that will be taken to achieve the aims i.e., tasks that will be done.

9 objectives aren't forbidden but later compress to 6-8. Bullet points again.

1. Research similar existing entry management and computer-vision systems.
2. Identify algorithms suitable for vehicle and passenger detection.
3. Research and develop a prototype interface using React for data visualisation.
4. Implement a camera module for automatic number plate and passenger detection.
5. Integrate federated learning to allow multiple gates to improve detection models locally while sharing model updates centrally, ensuring privacy and scalability.
6. Test the system under different lighting and outdoor conditions.
7. Evaluate the system's accuracy and reliability compared to manual logging.
8. Ensure scalability for multiple gates.
9. Document system performance, improvements, and the impact of federated learning.

2. Background and Context

Background

Provide the reader with a short summary of the relevant contextual information necessary to understand why your work is important. You should describe the context of your work (for example if it is relevant to a particular area of business, medicine, education etc) and how it addresses known issues in that context. This section is expected to have at least some references to external sources.

Summary of relevant published work (Literature review)

GUIDANCE: 1500 –2000 words

The literature review is a key part of your project and has specific marking criteria attached. The purpose of the literature review is to show that you have researched, understood, and evaluated all aspects of your problem statement. This is your 'investigation'.

You should describe the problem in detail, mentioning any organisations, institutions or individuals who are important in the context or the solution.

You should correctly reference key articles, books, or papers you have read that are relevant to the problem.

It is not enough to list relevant points. You also need to show the ability to:

- Summarise the main points.
- Discuss (e.g., point out strengths, weaknesses, differences of approach etc).
- Describe how the literature you have found relates to the problem you are addressing.
- Apply your own critical thinking to the material you include.
- Draw your own conclusions from your investigation and show how these will guide your methodology and implementation.

Everything that you write in your literature review must be clearly relevant to your problem statement and aims and objectives, and you must clearly state how this is the case.

You may include a table that summarises the benefits and limitations of the literature that has been reviewed.

Competitor Analysis/ Related work

This section should describe technical artefacts that are comparable to your project artefact. They might be openly available products in which case you may analyse things such as their usability, available features and price, or they may be research outputs. You may present some of this in a table.

First draft:

Paper	Problem Addressed	Data Used (Camera type, setup)	Methods & Models	Key Findings	Limitations & Issues	Relevance to my project
Nowruzi et al., 2019 —	Detecting passengers inside a vehicle using thermal images	• Thermal camera inside vehicle (FLIR) • Around 1284 images, augmented to around 25k • Controlled angle and lighting	• Lightweight, 4 layer CNN • Multi-task for CNN for classification and heatmap detection • Heavy data augmentation • Used Tensorflow Object Detection API	• Lightweight CNN performs comparably to larger models • Head detection is reliable for occupancy • Real-time is possible, edge-friendly	• Internal camera environment only • Small dataset • Not testing in outdoor or reflective conditions • Thermal distortions still exist (AC/heating)	High relevance for ML design and edge development, but limited to internal camera, my FYP focuses on external camera Useful for: model design, augmentation, head-based detection, small-model architecture
Paidi et al., 2019 — *Deep Learning based Vehicle Occupancy Detection in Open Parking Lot Using Thermal Camera*	Automatic detection of parking occupancy in harsh outdoor conditions (snow, low light, glare) Overcoming limitations of RGB cameras	• Axis Q1942-E thermal camera • Outdoor parking lot • 8007 labelled images • Thermal pseudo-colour frames (cannot show identity)	• Compared multiple object detectors • ResNet18 selected as best performer (96.16% AP) • ResNet50 underperformed due to dataset size	• ResNet18 best balance of accuracy + speed • Thermal camera effective in low-light, snow, and harsh lighting • Privacy-preserving by design (no identifiable visible features)	• Cannot perform ALPR (thermal cannot capture text/plates) • Cannot reliably count multiple occupants—thermal low detail. • IoU threshold 0.2 → low localisation accuracy.	Medium relevance. Shows strengths of thermal sensors but highlights they cannot meet FYP goals (ALPR + external occupant counting). Demonstrates real-world outdoor CV

	in Scandinavian environments		• Two-step pipeline: (1) detect vehicle, (2) map detection to fixed parking-template overlay	• Real-time capable.	• Template-based system not scalable—requires manual calibration. • High hardware cost for thermal imaging. • Only detects vehicle presence, not occupants.	issues (glare, weather). Motivates multi-sensor fusion: thermal for presence, RGB/NIR for occupants + plates. Reveals pitfalls to avoid: low IoU thresholds, template-based detection, overdependence on thermal imagery. Useful for: robustness insights, not for the full pipeline required in gate-entry system.
Malneedi & Soman, 2020 — *In-Vehicle Occupancy Detection And Classification Using Machine Learning*	Detecting the number of passengers and classifying each occupant as an adult or a child to control airbag	• Infrared Pi camera with wide-angle lens connected to a Raspberry Pi 3B microcontroller. • Camera mounted on the rear-view	• Haar Cascades for face detection (occupant counting). • Support Vector Machine (SVM) for adult/child classification.	• Achieved a face detection accuracy of 79.05%. • Achieved a face detection Recall of 90% and Precision of 86%. • Achieved a high adult/child	• System is limited to hatchback cars. Does not work properly when faces are exposed to bright light. • Future scope includes improving face detector accuracy and extending	

	deployment and prevent injury to children	mirror inside a hatchback car. 300 occupant images collected for training and testing. • Images of faces are resized to 190 x 190 pixels and converted to greyscale for classification.	• Image augmentation (rotation by 30 degrees) used to increase the children's face count for SVM training. • System implemented with a sequential pipeline of detection and classification.	classification accuracy of 99.36% using SVM. • The method is a low-cost and reliable system using machine learning.	to other vehicle types (Sedan, SUVs). • The classification method of a previous study failed when children held luggage or tightened their seatbelt.	
Ramik et al., 2013 — *Machine Learning Intelligent Vision for Autonomous Object Detection*	Creating a general-purpose autonomous system for object detection (this isn't specific to vehicles)	• RGB visual imagery • Uncontrolled environment • Real-world objects with clutter	• Visual saliency detection. • Object segmentation • Unsupervised + supervised ML • Hybrid distance metrics	• Strong performance in noisy, varying light environments • Emphasis on ROI extraction before recognition • Segmentation improves ML accuracy and efficiency	• Not about vehicles or thermal images • Not deep-learning based • Older methods compared to modern CNNs	Medium relevance Useful for: understanding pre-processing, ROI extraction, segmentation, illumination handling Helps design robust CV pipeline for an outdoor gate

Links for new literature review:

INDRA TECHNOLOGY

1. https://www.indragroup.com/cms-content/importacion/220711_pr_indra_vod_operating_on_three_express_lanes.pdf
2. https://www.indragroup.com/en/news/indra-centralizes-management-transurbans-roads-queensland-advanced-traffic-control-system
3. https://www.traffictechnologytoday.com/news/congestion-reduction/ai-based-vehicle-occupancy-detection-to-be-deployed-on-virginias-i-66.html
4. https://www.ibtta.org/sites/default/files/documents/2019/Lisbon/Patno_Brian.pdf This one has more companies mentioned, will go into this deeper
5. https://www.youtube.com/watch?v=3Fwl7x9WWnA

INVISION AI VEHICLE OCCUPANCY DETECTION

1. https://invision.ai/products/ivod/

ROADSIDE VEHICLE OCCUPANCY DETECTION SYSTEM (PATENT)

1. https://patentimages.storage.googleapis.com/0a/ec/5b/6c792afd5f65bb/EP4145404A1.pdf

NUMBER PLATE DETECTION AND RECOGNITION TECHNOLOGY

1. https://www.researchgate.net/profile/Shahnaj-Parvin-3/publication/350488454_Vehicle_Number_Plate_Detection_and_Recognition_Techniques_A_Review/links/65a50942c77ed9404779b0c4/Vehicle-Number-Plate-Detection-and-Recognition-Techniques-A-Review.pdf

HOT LANE DETECTION AND MANAGEMENT SCAN

1. https://rosap.ntl.bts.gov/view/dot/54893

TRAFFIC MANAGEMENT: MULTI-SCALE VEHICLE DETECTION IN VARYING WEATHER CONDITIONS USING YOLOV4 AND SPATIAL PYRAMID POOLING NETWORK

1. https://www.mdpi.com/2079-9292/11/17/2748
2.

Literature Review

Large temporary gatherings held on open fields often rely on multiple entry and exit gates to manage a moderate volume of vehicles and external workers through their designated gates. At the moment, these gates are typically managed with paper-based logs for vehicle registration numbers, passenger counts, arrival times, and gate number supported by radios for communication between staff. This arrangement makes it difficult to maintain a consistent picture of how many external

workers are on site at any given time, especially when not all volunteers have access to radios or digital devices. As a result, gaps appear in security and monitoring, such as missing entries for vehicles that have entered or left, inconsistent counting of external workers, and errors caused by rushed handwriting or miscommunication.

Several bodies of research suggest that a camera-based, automated approach could address many of these problems. Vision-based systems can detect vehicles, estimate the number of occupants, and capture licence plate information in real time, and this data can be passed directly into a central database rather than written by hand. Studies on passenger occupancy detection [1], computer vision for vehicle detection [2], and automatic licence plate recognition (ALPR) [3] show that it is technically feasible to replace manual logs with automated pipelines under a range of environmental conditions. At the same time, commercial solutions for vehicle occupancy detection on managed motorways [4], [5] demonstrate that similar systems are already being deployed at scale, although their internal workings are often not disclosed. The aim of this project is to draw on these bodies of work to design a non-invasive, camera-based artifact that records vehicle registration, time, passenger count in real time and supports better security and coordination across multiple gates at a large outdoor religious event, while keeping the event itself anonymous in reporting.

A recent review [1] of passenger occupancy detection methods provides a useful overview of how this problem has been tackled across different domains. The review [1] distinguishes between invasive approaches, such as sensors integrated into seats [6], seatbelts or weight-based systems [1], [6], and non-invasive approaches, which generally rely on external sensors such as cameras [5], radars [7], or Wi-Fi signals [1]. Invasive methods can be very accurate in controlled settings, for example in production vehicles where manufacturers can embed sensors into the interior [6], [8], [9], but they are unsuitable for uncontrolled environments such as temporary events, where vehicles are heterogeneous and organisers cannot modify visitors' cars. For this reason, the project focuses on non-invasive methods, with particular emphasis on camera-based systems that can be deployed at the roadside or gate without altering vehicles or passengers.

The same review [1] highlights the specific challenges of non-invasive occupancy estimation, especially when using cameras that view passengers through windscreen [10]. Factors such as changing lighting, reflections on glass, varying vehicle hights, and different searing positions all make it difficult to see occupants clearly from outside [1], [10]. In addition, privacy concerns arise when high-resolution RGB cameras capture identifiable facial information, especially in public or semi-

public settings. These observations are directly relevant to the project context: cameras placed at field gates will have to deal with bright daylight, glare, possible headlights at night, and a mixture of vehicle types, while also respecting the sensitivities of a faith-based event. The literature therefore motivates a design that combines robust pre-processing, for example, focusing on the windscreen region and mitigating glare, with lightweight models that work in real time. Other design ideas that may be considered are options such as infrared (IR) or near-infrared (NIR) imaging to balance visibility and privacy [2].

With non-invasive approaches, the review [1] notes that vison-based methods have become increasingly popular because they can provide both occupancy detection and contextual information about the vehicle, such as its class and registration plate. Alternatively non-visual methods such as radars, depth sensors [11], [12], or Wi-Fi signals can offer robustness under certain conditions, but they generally do not provide plate-level detail and are less straightforward to deploy cheaply at multiple temporary gates. This reinforces the decision to concentrate on camera-based techniques in this project, even though this introduces additional work in handling illumination, reflections, and partial occlusions. Overall, the survey [1] shows that while the general problem of occupancy detection has been studied widely, there is still a gap around practical camera-based systems that must operate at outdoor gates, handle diverse vehicles, and integrate seamlessly with licence plate recognition and centeralised event management.

Building on the general review, more specialised studies focus on vision-based detection of passengers and vehicles. Several works investigate in-vehicle camera set-ups [2], [9], [13], where a camera is mounted inside the cabin and observes occupants directly. Thermal imaging [2], [9] has been explored as a privacy option in this setting, because it reveals heat signatures instead of detailed facial features as shown in Fig. r1. In one such study [9], a small convolutional neural network (CNN) is trained on a modest but heavily augmented thermal dataset captured from a camera under the rear-view mirror as shown in Fig. r1, allowing it to predict both the number of passengers and their head locations in real time using low-power hardware. The results show that this lightweight model achieves around 91% exact counting accuracy and up to 99% binary (HOV-style) accuracy while remaining suitable for embedded deployment, and that head-focused detection is a strong cue for estimating occupancy in the cabin [9]. However, because these images are taken inside the vehicle (Fig. r2) with controlled viewpoints and without glass between the camera and occupants, they do not directly address the reflections and variable stand-off distances that are central in roadside gate scenarios.

Fig. r1 In-vehicle thermal image using FLIR One Pro thermal imaging camera [9]

Fig. r2 FLIR One Pro thermal imaging camera location for data capture [9]

Other in-vehicle systems use simple infrared cameras and classical machine learning rather than deep learning. For example, some work [6] mounts a Raspberry pi with IR LED camera (Fig. r3) near

the rear-view mirror, applies a face detector such as Haar Cascades to locate occupants, and then classifies cropped faces as adult or child using support vector machines. Despite relatively small datasets, such pipelines have achieved around 79% detection accuracy with an F1 score near 88% and adult/child classification accuracy of about 99% under constrained in-cabin conditions, and they are computationally cheap enough to run on modest embedded boards such as Raspberry Pi [6]. These findings are encouraging for this project because they show that a sequential pipeline, with detection followed by classification, can be effective, and that sophisticated hardware is not always required. At the same time, these systems are again designed for internal, forward-facing views, often limited to one vehicle type such as hatchback [6], and they report degraded performance under bright external light. In contrast, the proposed gate-entry system must cope with many vehicle styles, whose occupants will be seen through windscreens at varying angles, suggesting that more robust detectors (such as modern deep learning-based head or upper-body detectors) and careful handling od illumination will be necessary.

Fig. r3 Raspberry pi camera with IR LED [6]

Other exterior work uses thermal cameras and deep learning to detect vehicle occupancy at the parking-bay level rather than counting individual passengers. In one such study [2], a long-wave infrared camera mounted above an open car park was combined with several detectors, including YOLO-based single-shot models (average precision ~60%, log-average miss rate ~74%) and pre-trained classification networks such as GoogLeNet, ResNet18 (average precision 96.16%, log-average miss rate 19.40%), and ResNet50. The authors report that while YOLO-style models are fast, they struggle on low-texture thermal imagery, with coarser localisation and higher miss rates, whereas GoogLeNet and especially ResNet18 achieve much higher average precision and lower miss rates while still running close to real time (<1s per frame) [2]. This suggests that, for challenging visual conditions, backbone quality and feature extraction can matter more than raw speed, which is relevant to this project's choice of detector for outdoor gate imagery. At the same time, thermal images do not provide enough detail to read licence plates or reliably count individual occupants, and template-based mappings with low IoU thresholds can overstate true performance, reinforcing the decision to prioritise RGB or NIR cameras and to adopt more stringent evaluation metrics and flexible, non-template-based designs in the proposed system.

Fig. r4 Infrared image depicting a parking-lot [2]

Automatic licence plate recognition (ALPR) forms the second major component of the proposed system, because event organisers need to know how many people are in each vehicle as well as which specific vehicles have entered and when. Survey papers on licence plate detection and recognition [3], [14] describe typical pipelines in which a detector first localises the plate region and then a recogniser segments and decodes the characters, often using convolutional and recurrent neural networks to cope with variations in font, lighting, motion blur, and plate damage. Recent work has focused on making these pipelines robust in complex, real-world conditions while remaining efficient enough for real-time use on embedded or edge hardware, for example by combining lightweight object detectors with specialised recognition networks [3]. This research shows that automatic license plate recognition works reliably, so the project chooses to include an ALPR system to automatically record each vehicle's registration, entry time, and passenger count in a single digital log, replacing the need for manual paper records.

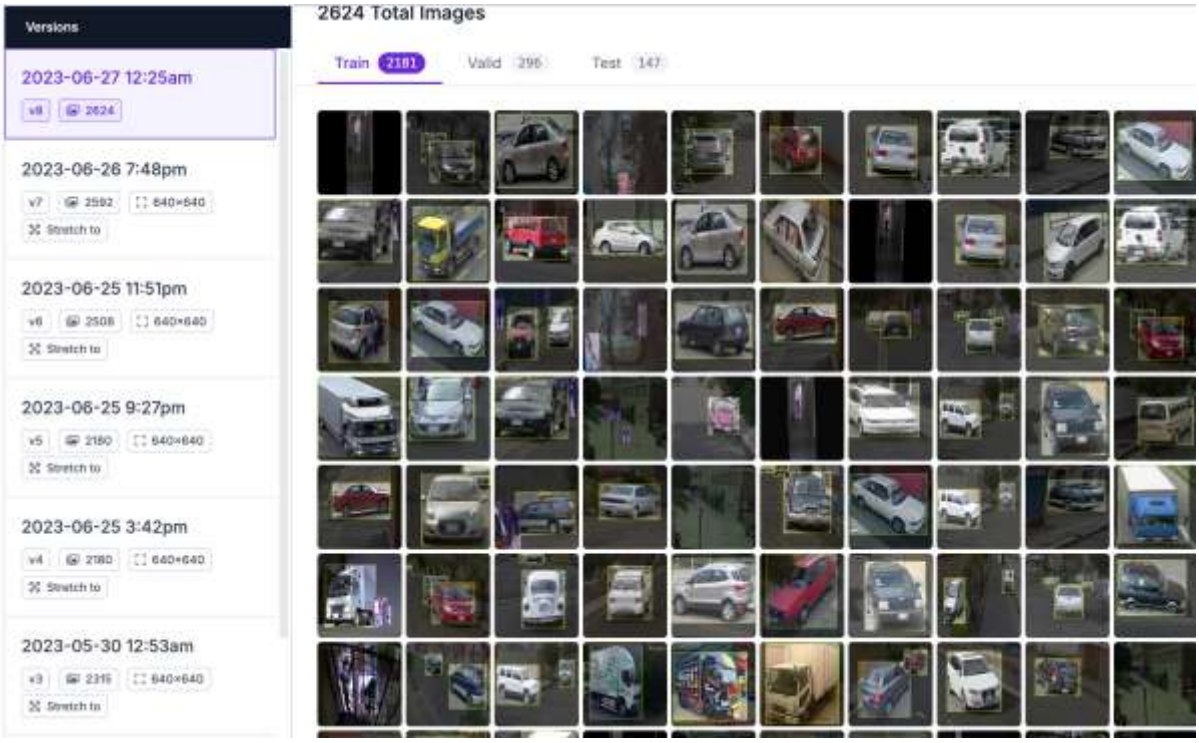
Alongside academic work, there are commercial roadside systems that claim to perform automated vehicle-occupancy detection for high occupancy vehicles (HOV) lanes in the United States. Vendors such as Indra [14] and Invision AI [5] advertise deep-learning solutions that can count front and rear occupants from a single roadside camera in real time, reducing manual works from enforcements. These deployments are important for this project because they demonstrate that combining occupancy estimation, plate reading, and centralised monitoring is practically achievable at scale, although the proprietary nature of the products means that details of their algorithms, hardware choices, and performance under different event-style conditions are not available. As a result, it is

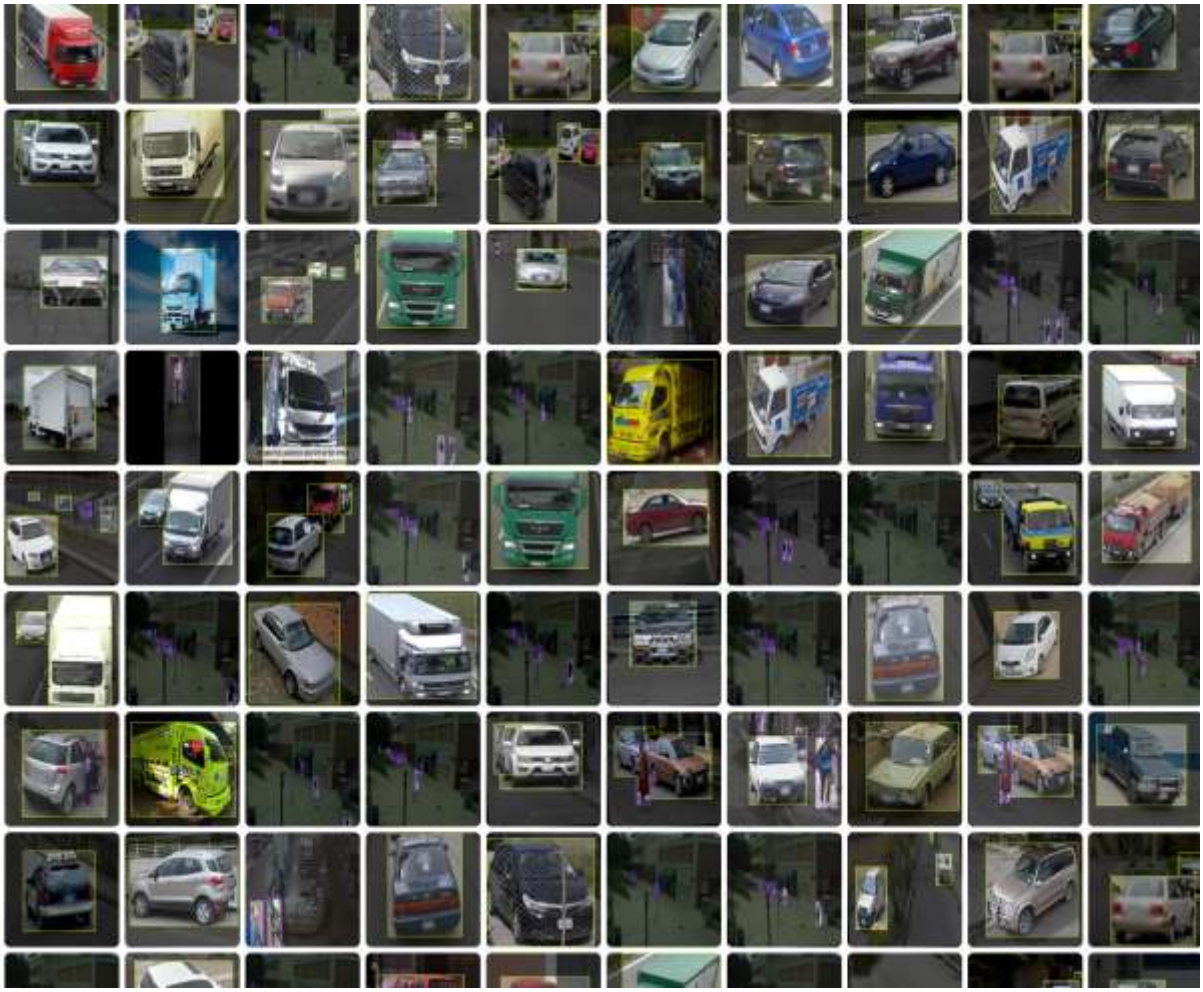
still useful to create and study an open, research-focused prototype for multi-gate field events, based on the same core ideas but fully documented and assessable in an academic setting.

3. Tools and Methods

The datasets I have researched and have landed in my top pick/s:

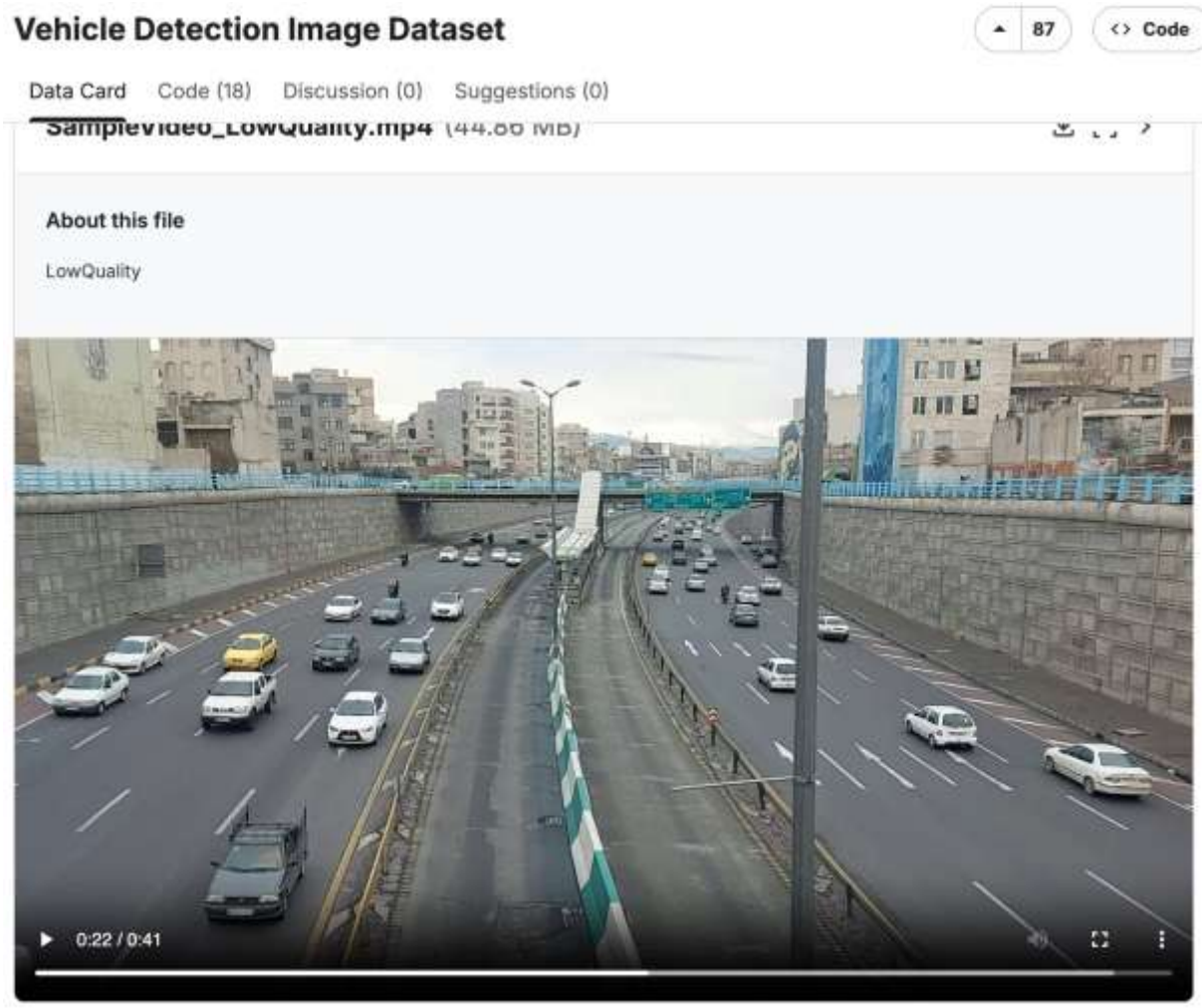
1. Surveillance for persons and cars
   https://universe.roboflow.com/jonac22/surveillance-for-persons-and-cars/dataset/8/images

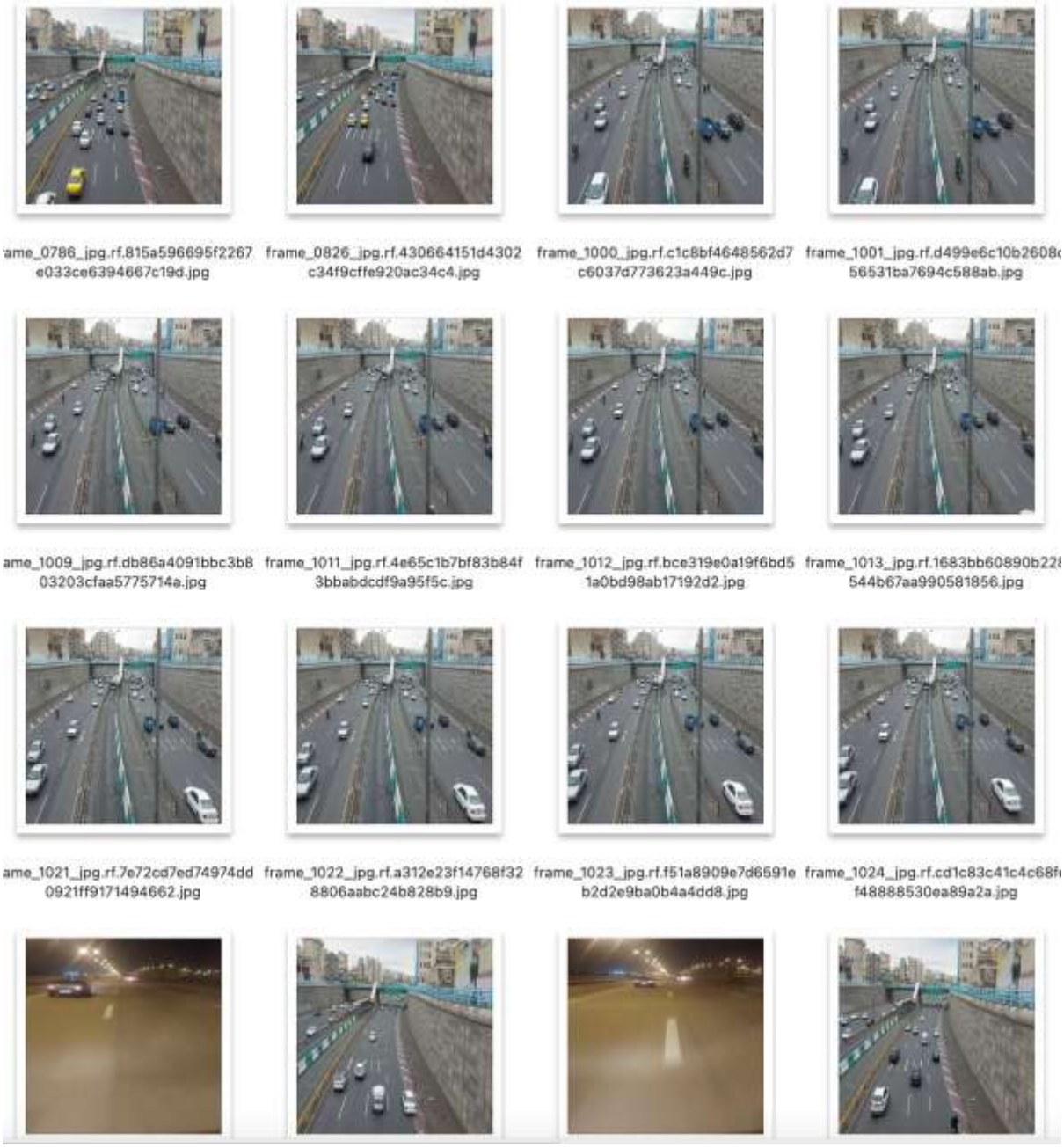




Advantages: Labelled, different environments, datasets combined together, different download formats depending on model, data is organised, different angles
Disadvantages: Car plates blurred, people are separate pictures and cars are separate, people inside cars are not visible, cars are mostly trucks

2. Vehicle detection image dataset
   https://www.kaggle.com/datasets/pkdarabi/vehicle-detection-image-dataset/data

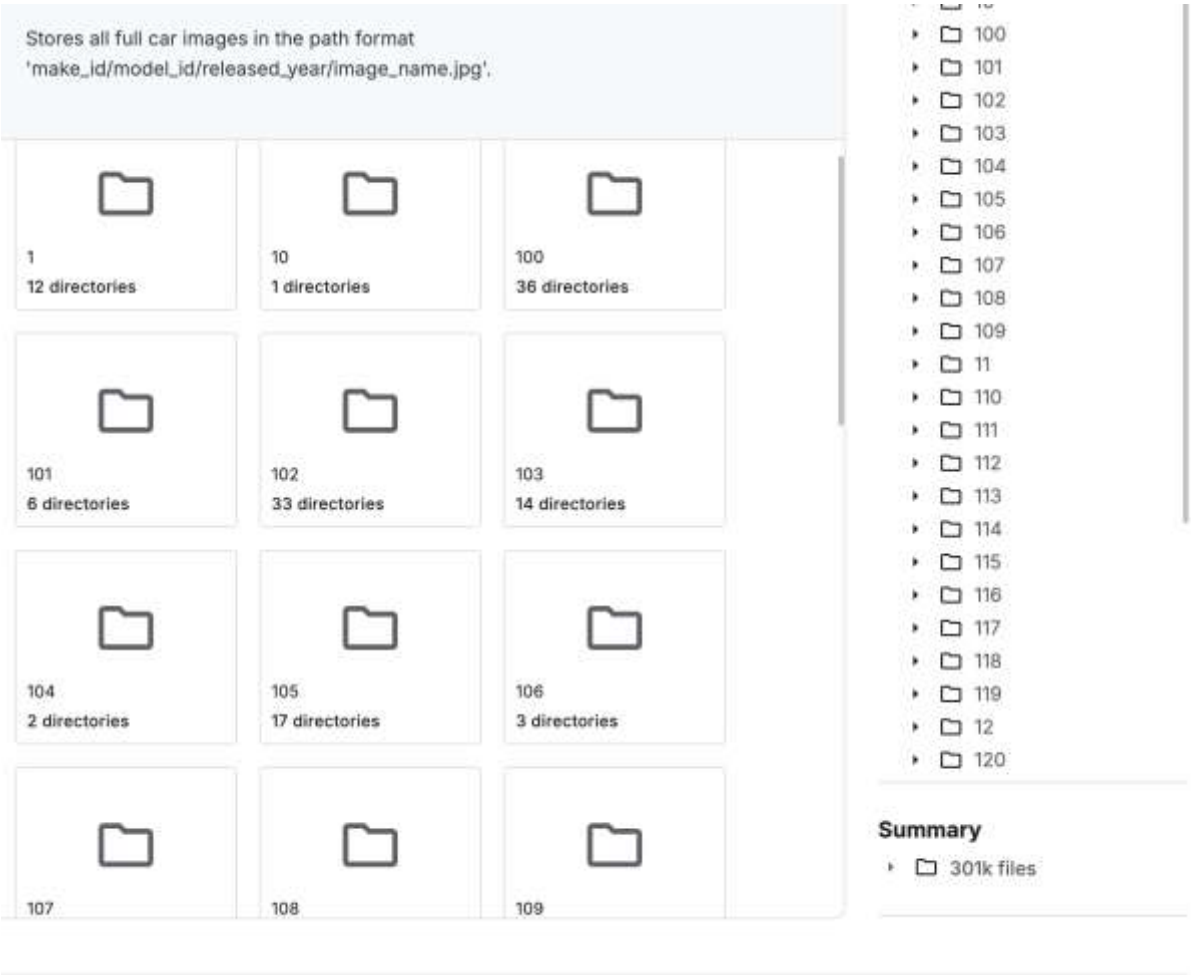

Vehicle Detection Image Dataset
87
Code
Data Card
Code (18)
Discussion (0)
Suggestions (0)
SampleVideo_LowQuality.mp4 (44.66 MB)
About this file
LowQuality
0:22 / 0:41



Advantages: video dataset but pictures are available in zip, clear view/high quality, organized, different filters and quality available, annotations.coco
Disadvantages: no people,, no car plates, unlabeled, one setting

3. CUHK CompCars Dataset

   https://www.kaggle.com/datasets/renancostaalencar/compcars/data


Stores all full car images in the path format
'make_id/model_id/released_year/image_name.jpg'.
1
12 directories
10
1 directories
100
36 directories
101
6 directories
102
33 directories
103
14 directories
104
2 directories
105
17 directories
106
3 directories
107
108
109
100
101
102
103
104
105
106
107
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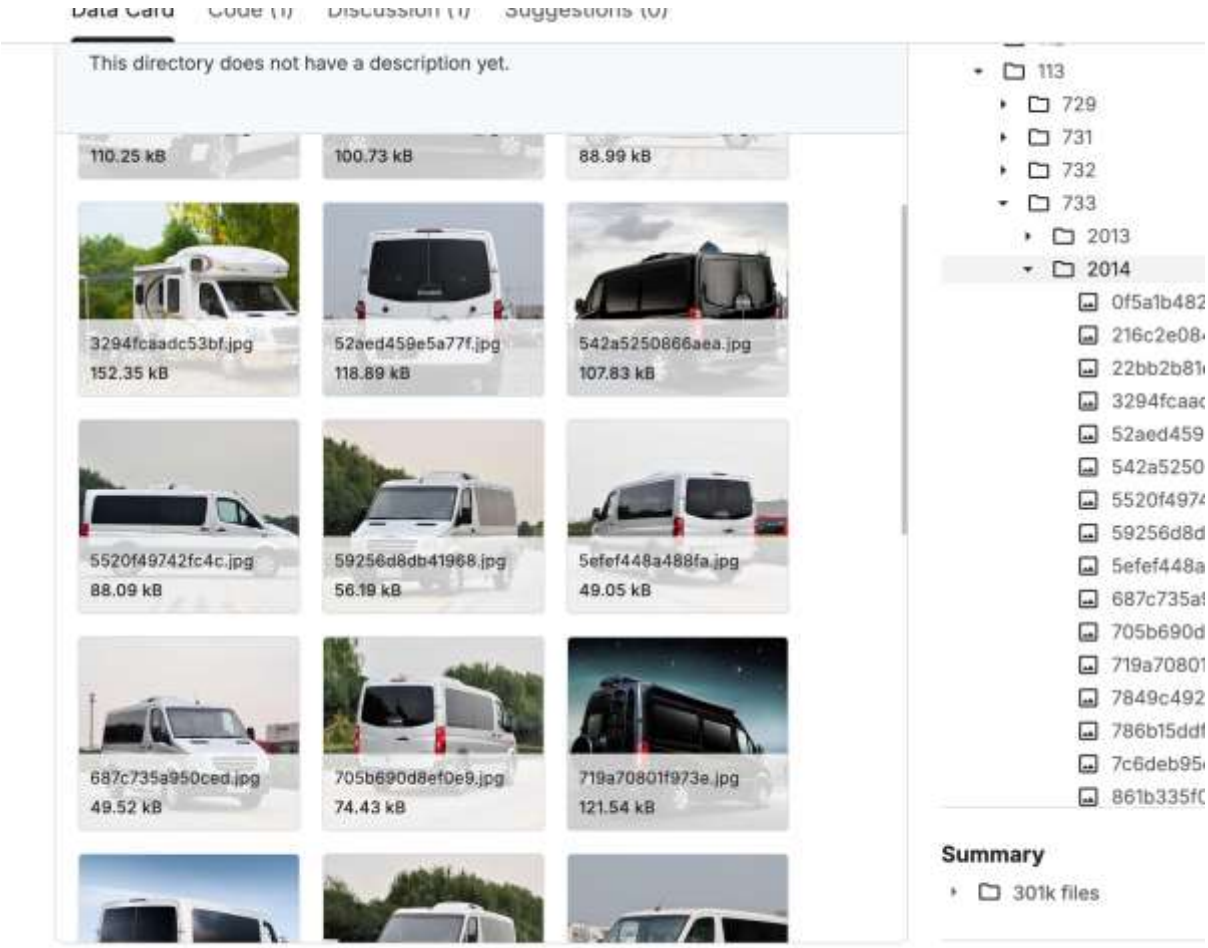
109
110
111
112
113
114
115
116
117
118
119
12
120
Summary
301k files



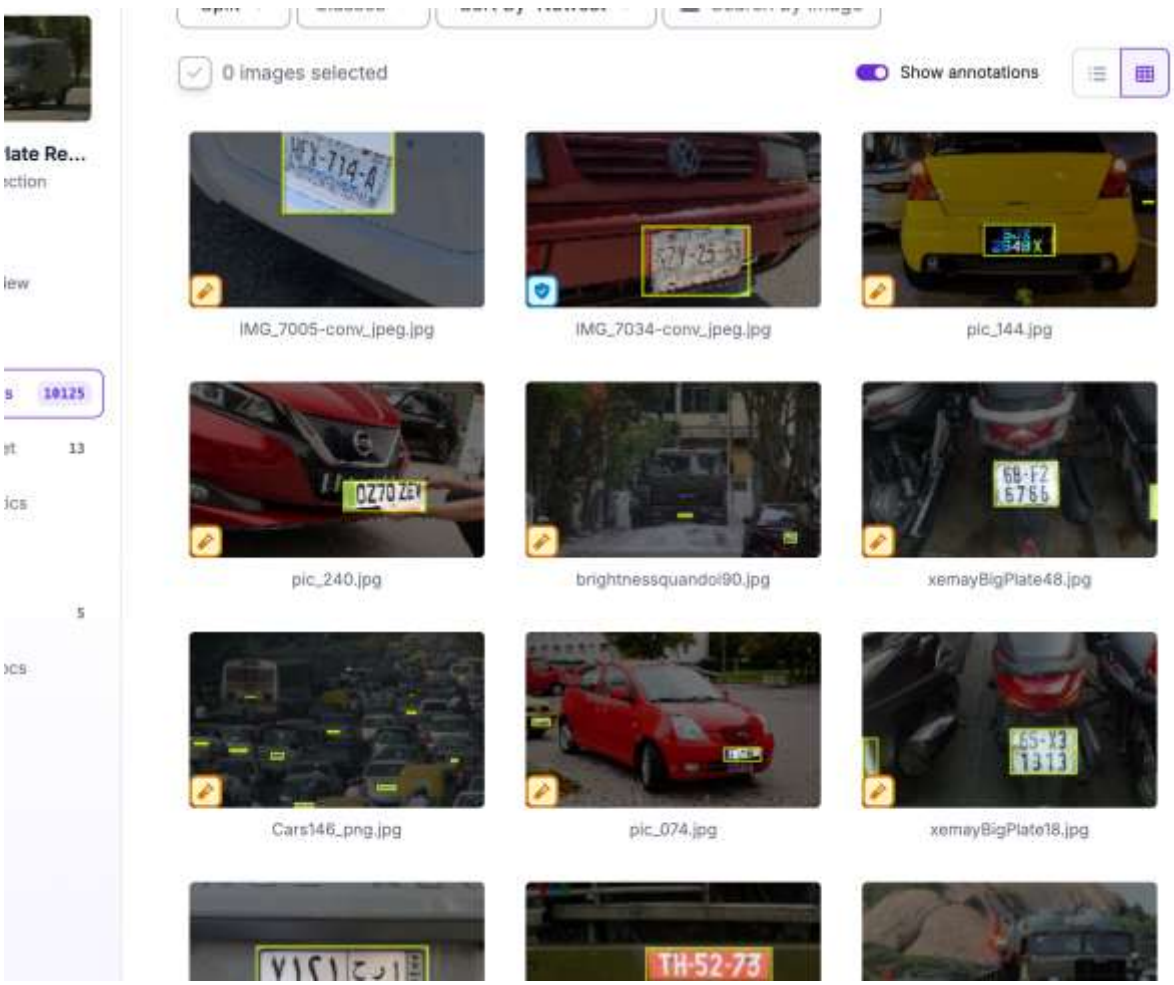
Advantages: large dataset, car models, different angles and lighting, some labelled

Disadvantages: no people inside the car, no car plate

The CompCars dataset features over 214,000 images across web and surveillance scenarios, covering 1,716 models labeled with viewpoints, parts, and technical specifications.

4. License plate recognition computer vison model

   https://universe.roboflow.com/roboflow-universe-projects/license-plate-recognition-rxg4e

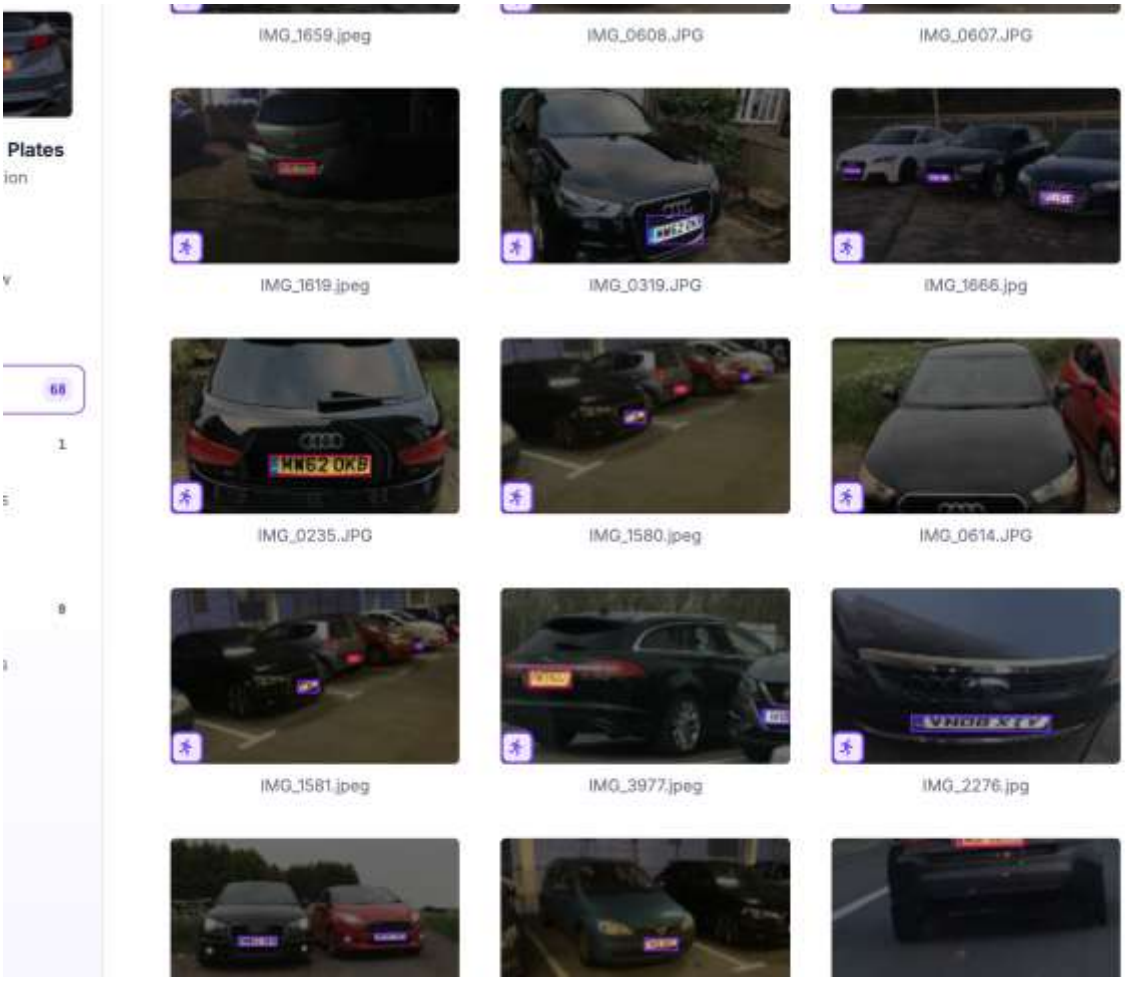


Advantages: car plate visible, labeled, different angles

Disadvantage: American car plates

5. UK license plates

   https://universe.roboflow.com/ryanpersonal/uk-licence-plates/browse?queryText=&pageSize=50&startingIndex=0&browseQuery=true



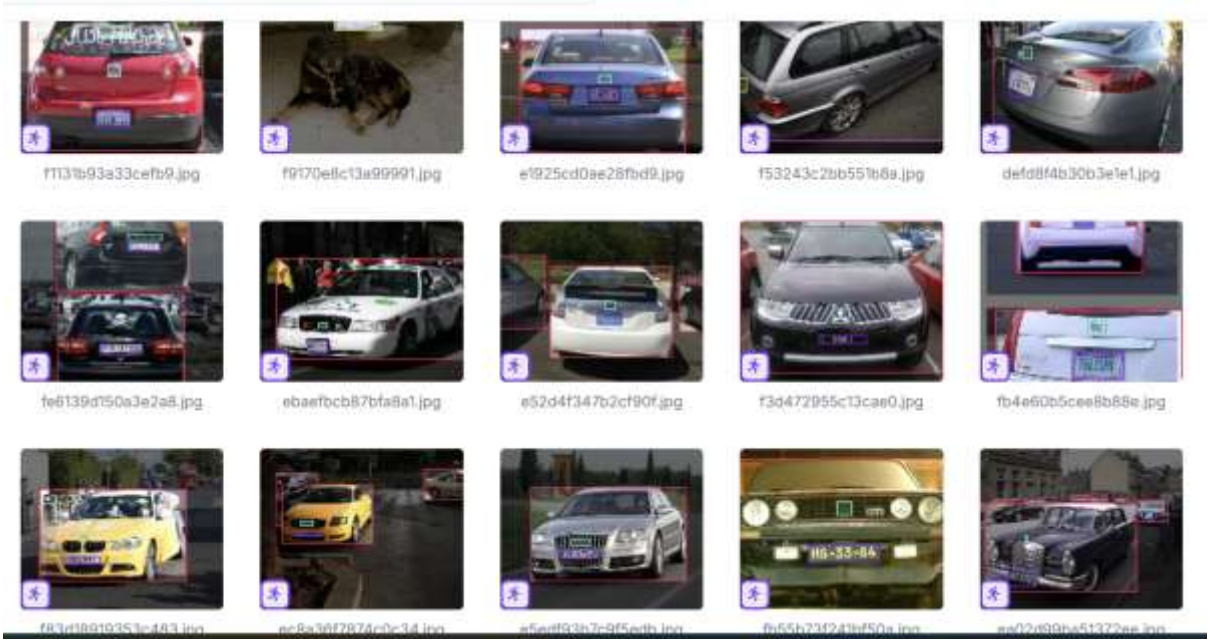
Advantages: UK license plates, clear images, different angles, annotated

Disadvantages: only 64 images, limited cars, limited settings and lighting (only a few dark images), no people in cars

Can combine more datasets with this

6. License plate detector computer vision dataset

   https://universe.roboflow.com/myworkspace-zkmzj/license-plate-detector-ozgvb



Advantages: different angles, different country plates, annotated, 834 images

Disadvantages: no people in the car, the pictures are very clear

7. DAWN dataset
   https://www.kaggle.com/datasets/shuvoalok/dawn-dataset

Fig. 4: Sample images of the DAWN dataset illustrating four cases of adverse weather conditions.

Advantages: different weather conditions, one lecturer worked on this dataset can ask for the resources, different angles
Disadvantages: no field settings, multiple cars instead of just one, car plates not visible, possibly out of scope

This section should answer the question -- **how you are going to undertake the project** and what factors influence your decisions.

Describe HOW you are going to create your artefact, including any tools, design methods, data gathering methods, algorithms etc that you are going to use. Tell us WHY you have chosen these methods in favour of others (with reference to the findings of your literature and technology review). Evaluate the impact of your decisions against the legal, ethical and professional issues you've identified.

Rationale for the technical tools selected

Write down which technologies – for example frameworks, software packages or libraries – you have used for the project and explain why they are suitable. Mention alternatives that you considered but rejected, with your reasons.

Rationale for the research methods used

Write about any methodologies you have used to conduct your project. For example, user centred design, agent-based modelling or different kinds of machine learning. Write about the methodologies in general eg. a brief history, what problems are they best suited for and some examples. Conclude with a paragraph about why this methodology was suitable for your project.

Project management and Product lifecycle

Write about project management tools you have used eg. github project, Trello, and the product lifecycle eg. agile that you have used to help you plan and manage your project.

Legal, ethical and Professional issues

All projects will have legal, ethical and professional considerations. You are marked on your ability to identify them and any actions necessary.

You must mention how these issues directly relate to your own project for example the data you have gathered or changes to the working practices of specific people.

Consider the issues that your project has under each heading. Discuss with your supervisor if you are not sure. Include the process you followed for ethical clearance (your ethics form should be included in an appendix at the end)

Risk register

Write down risks you have identified in your project and how you have mitigated them, for example by conducting proof of concept activities early, and/or conducting early stage user testing with lo-fi prototypes.. You may present this in a table if you wish. See resources at: https://moodle.roehampton.ac.uk/course/view.php?id=21620&sectionid=382313# .

4. Requirements/Specifications

Write down the specification or plan for your technical artefact, using industry-standard formats. We expect to see, for example:

- Personas
- User stories
- User journeys
- Use case diagrams
- Wireframes
- User flow diagrams

5. Solution design

This section should document the technical plan that you have used to create your artefact, for example

- UML class and sequence diagrams
- ERD diagrams
- network diagrams.

6. Solution implementation

Give a detailed account of how you have created your artefact. You are advised to divide this section up so that it matches your project plan, for example by describing the work you did in each sprint.

Consider choosing a particular technical challenge that you faced in each sprint and providing some detail (backed by evidence) of how you solved the challenge. Only include code snippets for particularly challenging parts of your implementation.

This section is all about your experience and therefore should be easy to write! Prepare for this by keeping a detailed log of all of your activities and using your github commits descriptively.

7. Solution testing/ evaluation

This is an important section where you weigh up the strengths and weaknesses of your artefact.

Guidance: If your project has a user-facing element, we expect to see some kind of evaluation of this with representative intended users, for example a 'think aloud' usability test.

You can also apply standard metrics for the domain you are working in and see how you have done against them. **Your project does not have to be perfect -- indeed the outcomes might have been bad.** The point is you must evaluate the outcome and discuss its strengths and weaknesses.

8. Discussion

This is a key section in which you summarise your findings, compare them to the findings of similar studies and draw conclusions about what your research has revealed.

Tie your discussion to specific evidence or activities that you have gathered and included in earlier sections of your report.

9. Conclusion

Summary

Briefly summarise your activities and findings.

Recommendations for further work

In this section, list some activities that you would undertake to improve your project if you had more time. Make sure these are realistic and based in evidence from your evaluation section.

Reflection

Finally, critically reflect on the entire project process and how well you have worked on the project. What particular things have you learned during the project? Why were you able and unable to meet project goals? What would have you done differently in hindsight? Write this section in the first person.

10. References

In this section, you **must** reference any sources used in your work. Typically, these sources will have come up during the investigation and related work sections. Your referencing must use the IEEE referencing style IEEE Citation Guidelines2.doc (ieee-dataport.org) .

It is **highly** recommended that you use reference management software such as RefWorks that is provided by the university. Your project should have as many references as is required. However, having few references indicates that no thorough investigation has occurred.

It is your responsibility to ensure that you have actually read all the material you reference, and that the references provided in your report are legitimate and **NOT AI generated**.

11. Appendices

Appendices appear after references. Your appendices depend on the nature of your project. **Do not assume people will read your appendices.** Even if you direct them to do so in your main text, appendices are considered additional information and should not be relied upon to understand your main body of work. Refer readers to an appendix using a phrase such as *see Appendix A for further details*.

The following documents **must** be included as references:

- Your Project Proposal poster
- Evidence of your use of a project management tool eg. screenshots
- A description of how to access any technical output. **It is strongly recommended you use GitHub or something similar to do this.**
- Your ethics form
- Your risk register
- Any important communications between you and external stakeholders -- **please ensure private data is removed and communications anonymised.**